COVER PAGE

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Introduction

This report will present a comprehensive analysis of the “fraud_victim_dataset”, with theprimary aim of identifying fraud patterns and developing data drtiven strategies in order to mitigate that fraud. This report is divided into five sections: Data Understanding, Data Cleaning and Preparation, Exploratory Data Analysis (EDA), Fraud Analysis, and Strategic Recommendations for management.

Data Understanding

This section will identify the different variables and their types in the dataset and also present a summary of the key statistics.

The analysed dataset has 500 rows and 23 columns. There are 23 variables, which contain a mix of categorical data (for example, gender) and numerical data such as “transaction amount, which is a financial metric.

Table 1: Variable overview

Variable	**Variable Type**	**Description**
Transaction ID	Categorical	Unique ID for each transaction
Customer ID	Categorical	Each transaction is connected to a customer ID, each customer has a unique ID
Age	Numerical	Age of the customer
Gender	Categorical	Male/Female
Education	Categorical	Education level - Bachelor/High School etc
Marital Status	Categorical	Married, Single, Divorced
Region	Categorical	North, East, West, South
Personal Income	Numerical	Annual individual earnings
Family Income	Numerical	Annual family earnings
Credit Score	Numerical	Credit score

Loan Amount	Numerical	Loan amount
Has Loan	Categorical - Binary	1=Yes 0=No
Owns Home	Categorical - Binary	1=Yes 0=No
Transaction Amount	Numerical	Amount spent
Transaction Type	Categorical	Online/In store etc
Device Type	Categorical	Device used for transaction
Account Age Days	Numerical	How long (days) the account has been open
Number Previous Transactions	Numerical	Number of previous transactions made
Average Transaction Value	Numerical	Average value of previous transactions
Internet Usage hours	Numerical	Daily internet usage (hours)
Mobile Usage Hours	Numerical	Daily mobile usage (hours)
Is Fraud	Categorical - Traget	1=Yes 0=No
Fraud Amount	Numerical	Monetary value

Table 1 gives an overview of all 23 variables in the dataset, including the data types of each variable and a description.

Table 2: Age descriptive statistics

Age	
Mean	50.1074
Standard Error	0.64485503
Median	50.2
Mode	51
Standard Deviation	14.4193968
Sample Variance	207.919003

Kurtosis	-0.2455998
Skewness	0.15838289
Range	72
Minimum	18
Maximum	90

The customer ages range from 18-90. The mean, mode and median are very similar (around 50 years), indicating a mostly symmetrical distribution. The standard deviation suggests that there is a moderate spread around the mean.

The mean transaction amount is 2004, while the median is 1375. This indicates a non symmetrical distribution of data. The standard deviation of 2095, which is high, suggests that there is great variance in this data and many outliers present. This variance is further highlighted by the range of 10-15040.

Data Cleaning and Preperation

This section will outline the steps taken to clean the data and prepare the data for EDA and further regression analysis.

With the use of Excel's filtering system, each variable was inspected and no blanks or null values were detected in this dataset, hence no adaptations were required.

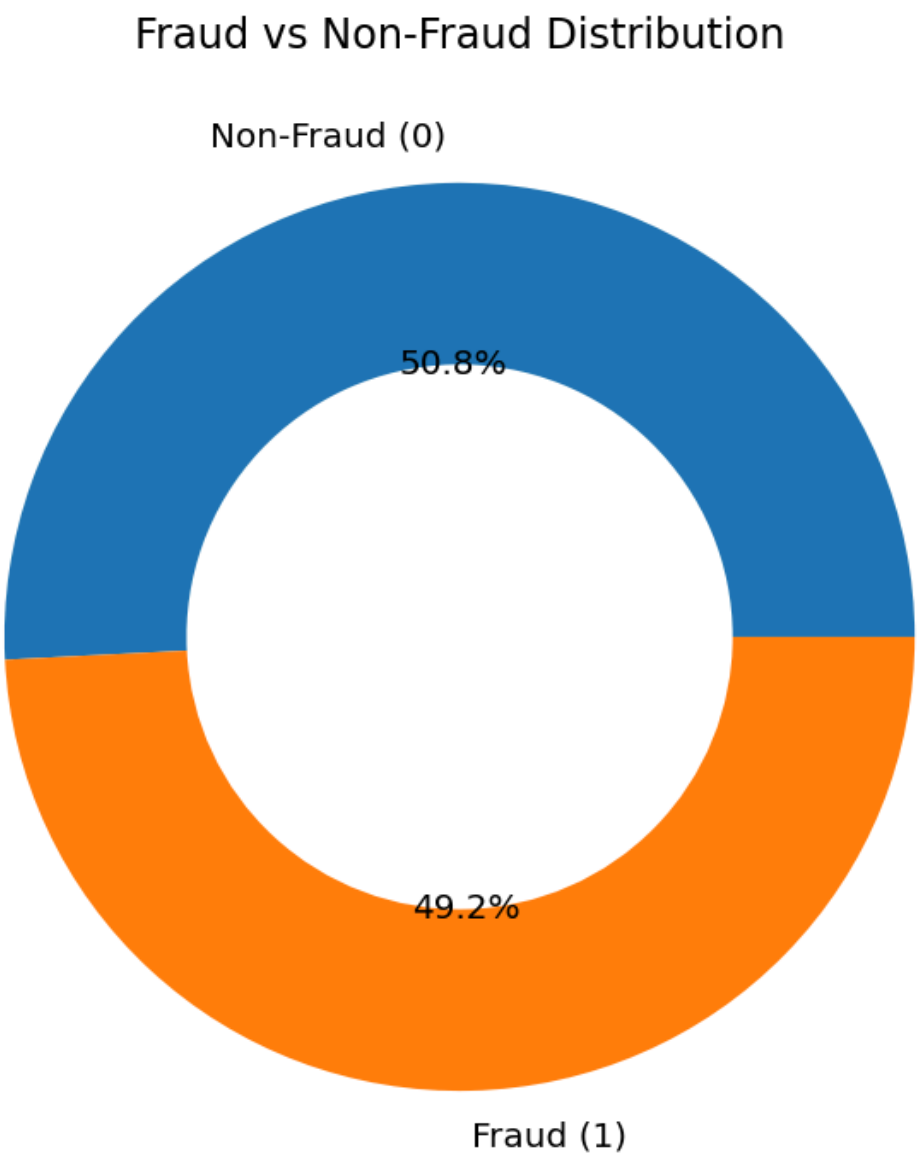
The age variable contained decimals, for example 51.1. In order to ensure accuracy these figures were rounded to the nearest whole number with the use of Excel's "Round()" function.
By analysing the "transiction_amount" outliers were detected. The mean transaction amount is 2004 and transactions above the value of 10000 were flagged as "Extreme" in a new separate column "outlier_flag" for further analysis.

In anticipation of regression analysis, variables such as age, transaction amount, credit score and personal income were standardised with the Excel Standardise function.

Exploratory Data Analysis

This section will present different visualisatons (histograms, bar charts etc) made with 'Python'. These visulaisatios will be used to analyse trends, distributions and correlations between variables.

Fig**ure 1: Fraud vs Non-Fraud Distribution**



In this dataset, the amount of fraudulent transactions (49.2%) are similar to legitimate transactions (50.8%). This is an unusually large amount of fraudulent transactions.

Figure 2: Transaction Amount Histogram

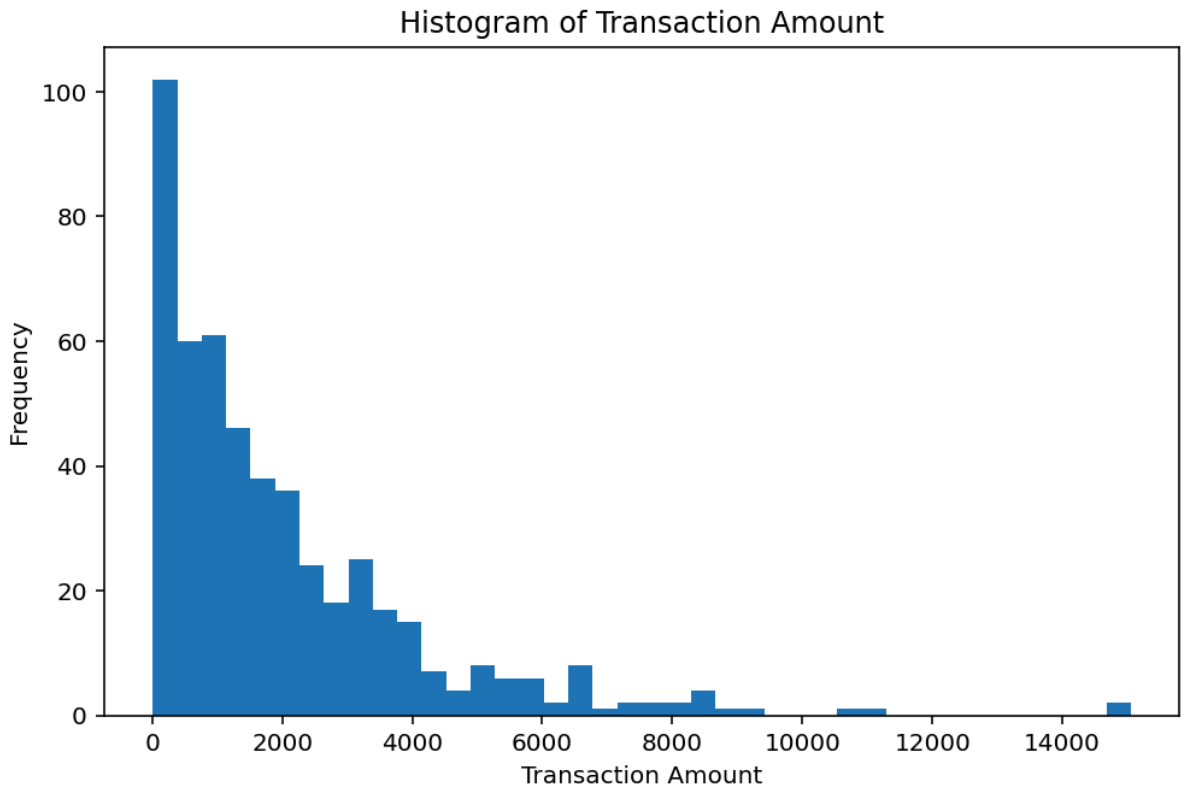
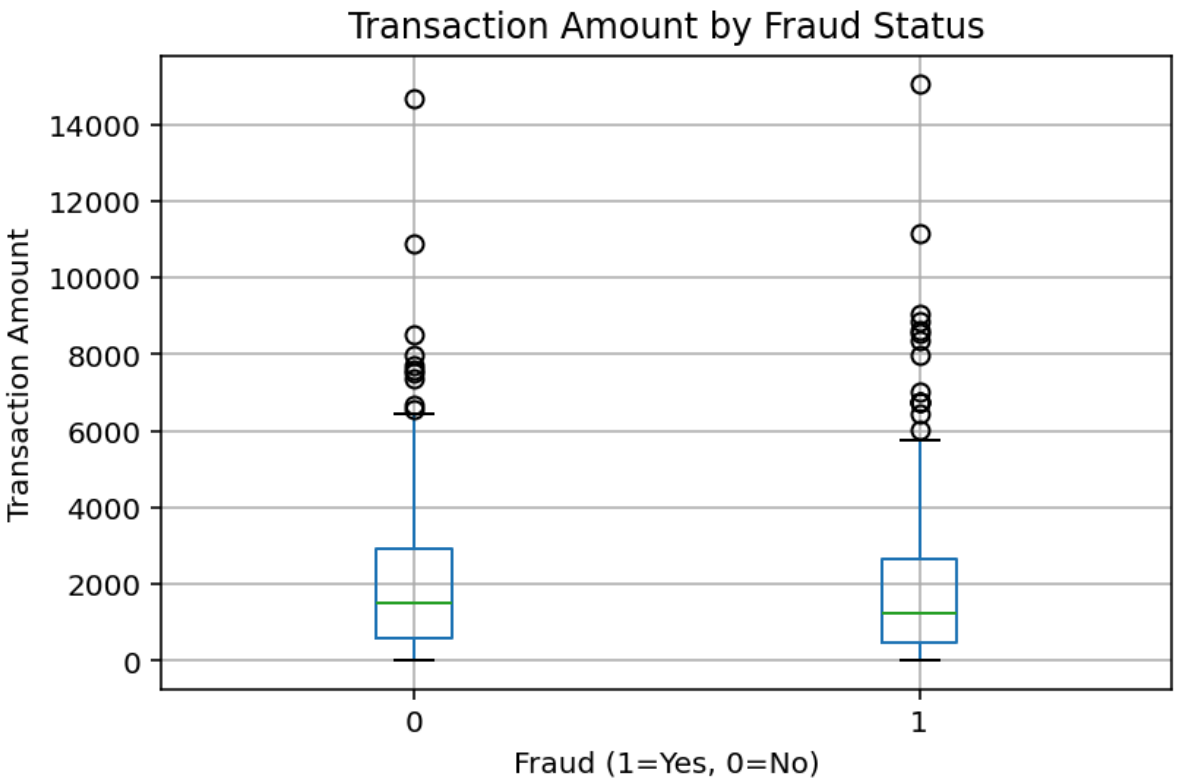


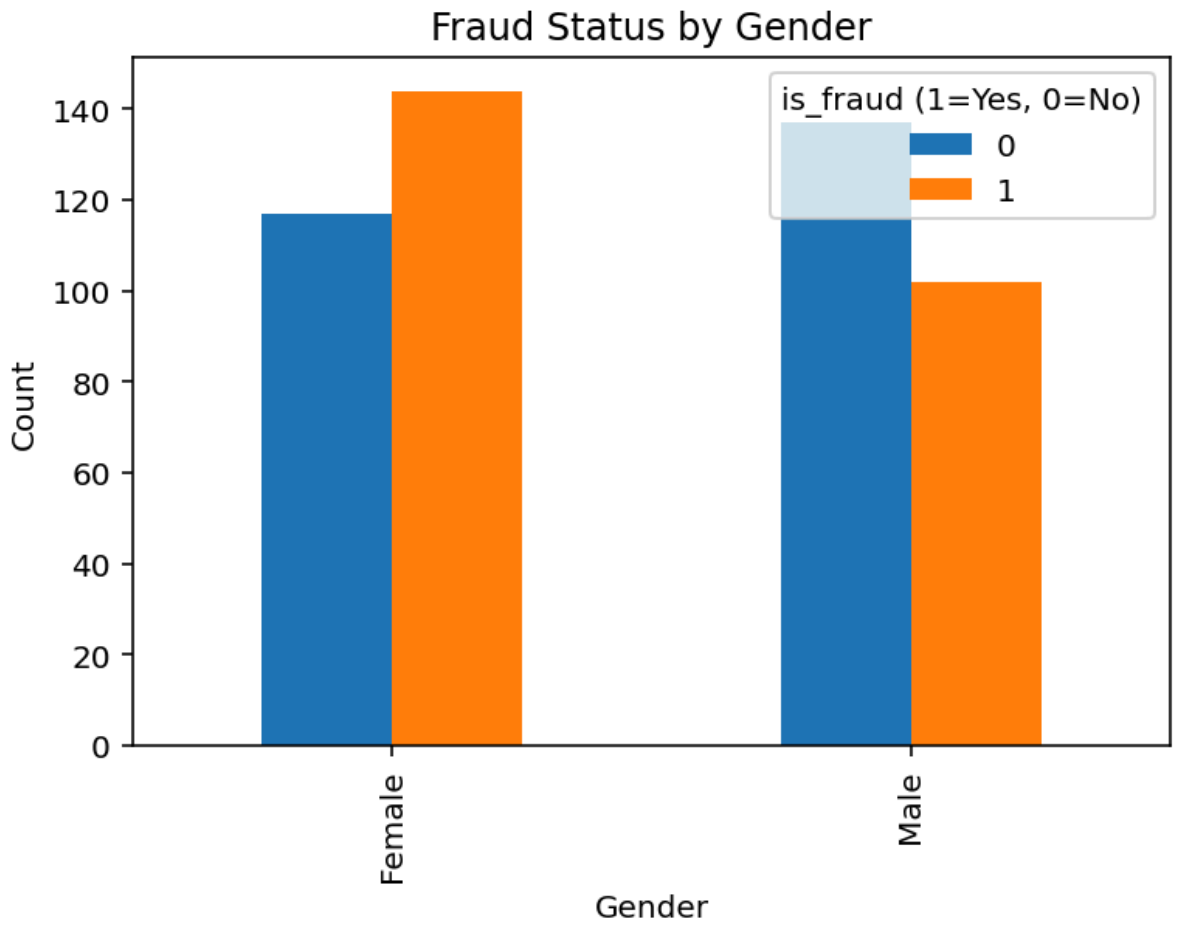
Figure 1 shows that the distribution of transaction amounts is right-skewed,showing that the majority of transactions are of small values, making it easier to see the outliers. Larger amounts of over 10,000 which are not frequent, these unusual patterns potentially indicate fraud.

Figure 3: Transaction amount box and whisker plot



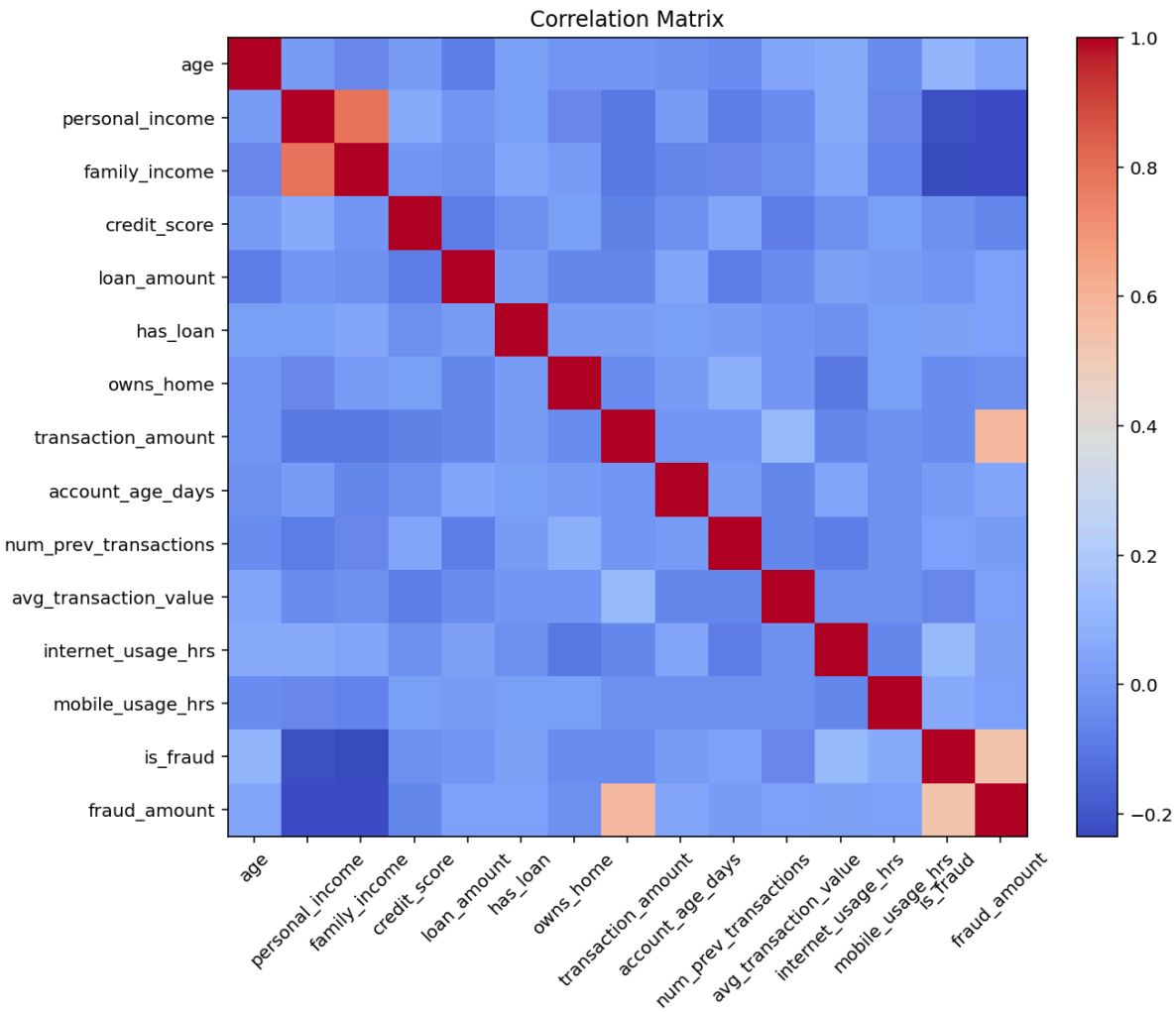
The distribution for both fraudulent and non-fraudeulent transactions are mostly similar, as is the Interquartile range (represented by the blue box) for both, this indicates that the value of fraudulent transactions are similar to that of normal customer spending in most cases. The median transaction amount (green line) for fraudulent transactions is similar to that of non-fraudulent transactions. Overall Figure 2 shows that transaction amount is a weak indicator for fraud detection.

Figure 4: Fraud status by gender bar chart



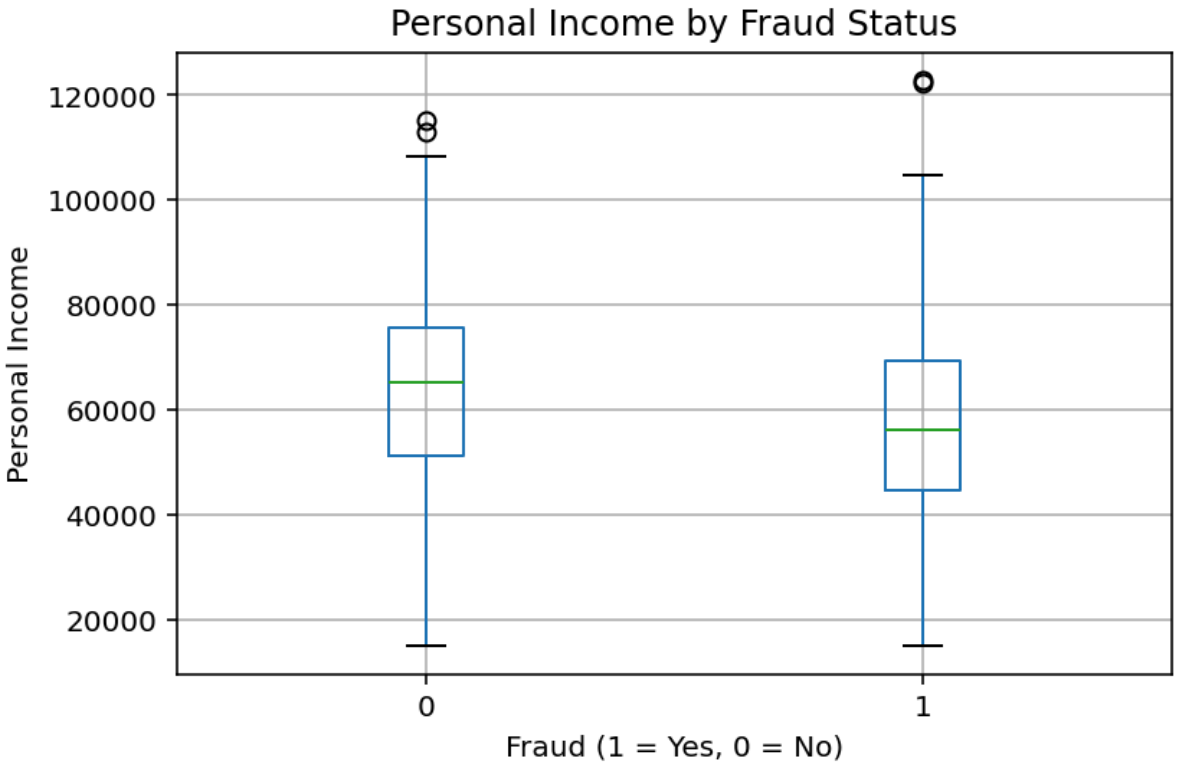
Amongst female customers the amount of fraudulent transactions excdees the legitimate transactions, whilst among male customers the reverse is applicable. This data shows us that female customers are associated with a higher rate of fraud than that of male customers. Gender could be a target for fraudulent activity.

Figure 5: Correlation Matrix



There is a strong negative correlation between fraud and both personal and familial income, the lower income is the more prevalent fraudulent transactions are. This means that in this dataset low income customers are associated with fraudulent transactions more than high income customers.

Figure 6: Personal income by Fraud Status box and whisker plot



There is a downwards shift in distribution, where fraud is present. The median (green line) personal income is lower in fraudulent cases than non-fraudulent cases. This reinforces the trend of low-income customers being susceptible to fraud.

To summarise, during the EDA it was found that transaction amount is not a good indicator for fraudulent transactions, nearly half of the observed transactions are fraudulent and that female and low-income customers are more suseptcile to fraud.

Fraud Analysis

In order to identify anomalies in the dataset, the use of Z-Scores were applied to the ‘transaction_amount” variable. The Z-Score for each transaction amount was calculated using the formula ‘Z=X−μ/σ’, where X= chosen transaction, μ= 2004, which is the mean of transaction amounts and σ= the standard deviation which is equal to 2095.

The Z-Score in this case signifies how many standard deviations any given transaction amount is from the mean. A Z-score of greater than 3 is unlikely under normal circumstances and therefore transaction amounts that had a Z-Score greater than 3, were marked as “high risk” for potential fraud.

Figure 7: High risk transactions

	A	B	C	D	E	F	G	H	I	J	K	N	O	P	Q	R	S	T	U	V	W	X	AB	AC
1	ansactio	ustome	age_cle	gende	educati	arital_st	regio	rsonal_ir	mily_inc	redit_sc	oan_am	action_	Outlier_l	nsaction	device_t	ount_ag	prev_tra	ransacti	rnet_usi	bile_usa	is_frau	aud_am	Z_scor	Fraud_F
25	T00024	C9383	29	Male	Bachelor	Widowed	East	37464.65	79910.64	560	56563.78	8562.81	Normal	In-Store	POS	4647	169	6857.55	1.7	6	1	5795.27	3.129203	High Risk
44	T00043	C2005	48	Female	Master	Widowed	North	15000	53580.86	641	6579.38	11158.44	Extreme	ATM	Mobile	3379	43	8359.47	3.2	5.1	1	10218.89	4.367576	High Risk
130	T00129	C8197	52	Female	High Schoo	Married	West	44617.15	82850.41	547	10738.02	8343.65	Normal	ATM	ATM	3205	61	5198.78	4.5	5.4	1	7597.73	3.024642	High Risk
147	T00146	C7445	62	Female	Bachelor	Single	North	50167.28	63203.95	653	51132.59	8848.03	Normal	Online	POS	1915	125	4705.08	4.3	8.8	1	4994.73	3.265281	High Risk
331	T00330	C3980	59	Male	Master	Married	North	50958.19	59968.6	772	41667.17	9042.64	Normal	In-Store	POS	1453	112	8266.27	3.1	9.3	1	5484.63	3.35813	High Risk
337	T00336	C5757	48	Female	High Schoo	Single	West	55059.49	86912.7	708	23444.5	15050.41	Extreme	In-Store	Mobile	1274	41	5868.73	6.8	10.5	1	8190.71	6.224432	High Risk
383	T00380	C1131	38	Female	Master	Single	East	38578.93	59105.26	744	36508.87	8620.45	Normal	ATM	Mobile	3205	234	4052.4	0.5	2.3	1	7091.7	3.156703	High Risk
502																								

Using this method, these 7 transactions were flagged as “high risk” for potential fraud and require further review.

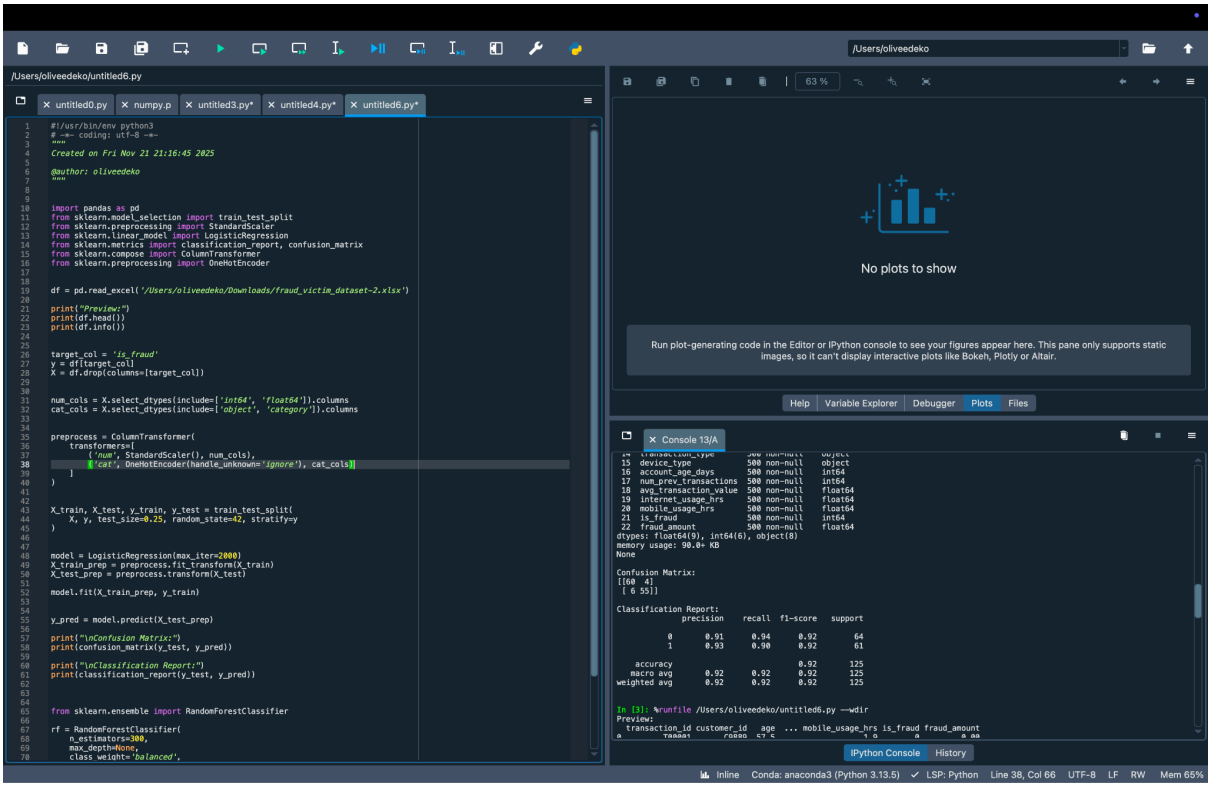
Table 3: Regression Output

Variable	Coefficient	P-value	Significance
Intercept	0.7297	0.000	Significant
Transaction Amount	-0.00001	0.176	Not Significant
Age	0.0038	0.013	Significant
Credit Score	-0.0001	0.709	Not Significant
Personal Income	-0.000005	0.000	Significant

Regression analysis was carried out on Excel using “is_fraud” as the target variable and Transaction Amount, Age, Credit Score, Personal Income as predictor variables. By interpreting the p-values, it can be concluded that:

- Older customers are likely to be victims of fraud.
- Customers with higher income are less prone to fraud.

Figure 8: Python code



On Python a logistic regression model was trained with the “fraud_victim_dataset”, the dataset was split into training (75%) and testing (25%) subsets, and model performance was then evaluated using the test set.

Table 4: Confusion matrix

	Pedicted Non-Fraud	**Predicted Fraud**
Actual Non-Fraud	60	4
Actual Fraud	6	55

Table 5: Classification Report

Metric	**Non-Fraud**	**Fraud**
Precison	0.91	0.93
Recall	0.94	0.90
F1-Score	0.92	0.92

The logistics regression model had an overall accuracy of 92%, it identified the majority of fraud and non-fraud cases correctly.

High precision for fraud, 0.93 shows that when the model predicts fraud, it is highly accurate
High recall for fraud, 0.90 indicates the model successfully identifies most fraudulent cases.
The balanced F1-scores, 0.92 show that the model performs at a consisten level across fraud and non-fraud categories.
A small number of false negatives (6 cases) is an area for concern, as undetected fraud has greater negative impact than false alarms.

Discussions and Recommendations

From the exploratory data analysis and the fraud analysis carried out on this dataset, it can be concluded that older customers and customers in lower income brackets are prone to fraud.
To mitigate fraud and improve customer satisfaction the company should apply the following recommendations.

Implement SAS's Agentic AI

This AI detects fraud and makes key decisions autonomously. It analyses suspicious transactions and evaluates contexts, makes a decision on whether to block or approve the transaction without human input. This AI "integrates multiple layers of analytics, combining machine learning and real-time decisioning to deliver a multi-dimensional defense" and can detect and stop fraud within miliseconds. (Lykourina, 2025)
This is a proactive appraoch that would reduce fraud and improve the satisfaction of customers.

Customer Education Programmes

Programmes tailored to vulnerable customers that the analysis identified, such as older customers and low-income customers. These programmes can assist customers by educating them on safe internet usage, fraud tactics and warning signs of fraudulent behaviour. If vulnerable customers are more aware of these factors and can successfully mitigate fraud, customer satisfaction will be improved greatly.

Conclusion

Analysis of the dataset identified older and low-income customers as targets for fraud. To mitigate fraud and improve customer satisfaction, it is recommended that the organisation implements a proactive approach to fraud detection and educates vulnerable customers on how to identify and protect themselves from fraud.

References and Bibliography:

- Lykourina, A. (2025) 'The future of trust: AI is powering a new era in banking fraud detection', *SAS Blogs*, 21 November. Available at: https://blogs.sas.com/content/sascom/2025/11/21/the-future-of-trust-ai-banking-fraud-detection/ (Accessed: 22 November 2025).